

PEDIATRIC CARE



Child 6 mos -6 years
Barky cough
And/or Stridor
***Refer for In Person
Healthcare visit for stridor to
triage**

Indications for face-to-face visit
and exclusions from pathway

- Pre-existing upper airway abnormality
- Hypotonia or neuromuscular disorder
- Recent airway procedure or intubation

- Parental report of ill or toxic appearance
- Airway obstruction
- Impending Respiratory failure
- Altered mental status
- Excessive drooling, gasping for air, severe retractions, tachypnea, grunting, head bobbing, see-saw breathing, cyanosis

Yes

- Exit pathway
- Call 911
- Assess and Stabilize
- Initiate ALS
- Transport to higher level of care

No

Mild Croup

Barky Cough with any of the following:

-No stridor at rest

- Stridor only when agitated
- No respiratory distress
- No mental status changes
- No or minimal retractions
- No tachypnea

Ok to treat virtually or face to face

**One dose of Dexamethasone
Oral, IV or IM (0.6 mg/kg)
Max Dose 16 mg*

Moderate/Severe Croup**Stridor at rest** with/without any of the following:

- Moderate to severe retractions
- Moderate to severe tachypnea
- Grunting, head bobbing, see-saw breathing
- Decreased air entry
- Anxious or agitated or tired appearing

Face to face visit evaluation recommended

**One dose of Dexamethasone Oral, IV or I
M (0.6 mg/kg) Max Dose 16 mg
***Nebulized Racemic Epinephrine 2.25%
0.5 mL
Depending on severity, consider activation
of EMS (call 911)

Reassess in 30 mins

Clinically
Improving?

Yes

Observe for 2 hours
(Depending on local
resources)

No

Recurrence of
moderate to severe
symptoms?

No

Discharge home once
discharge criteria is met

- Re-dose nebulized racemic epi and
- Transfer for higher level of care (ED, Inpatient, PICU)
- ****Consider alternative diagnoses

**Pediatric Transfer
and Advice Center:**
562-658-3980 OR
8-137-3980

ED/
INPATIENT/
PICU

Discharge Criteria:

- No or minimal retractions
- No stridor at rest
- Tolerating orals
- Parents able to return pm
- <=2 doses of Racemic epinephrine given after Dexamethasone

Not Routinely Recommended

(No evidence suggesting the use of)

- Chest/Neck X-ray
- Antibiotics
- Viral Studies
- Cool mist in hospital/ED setting



CROUP SUMMARY AND FOOTNOTES

Croup is a viral respiratory illness that causes inflammation of the larynx and subglottic airway. It is characterized by symptoms of stridor, barking cough, and hoarseness.

These guidelines are based on current evidence and consensus to assist with clinical management and decision making.

FOOTNOTES:

****Dexamethasone^{1,2,3,6,-}**

- Oral and IM forms are equally effective, IV form has shorter duration of action
- Oral 1mg/ml syrup is unpalatable, can consider giving IV form orally instead
- Onset of glucocorticoid effect in one hour, found to reduce symptoms of croup at 2 hours
- Max dose 16 mg⁶

*****Nebulized Racemic epinephrine^{4,6}:**

- Clinical improvement can be noted within 30 minutes
- Standard dose of 0.5 mL
- Can repeat the dose after 30 minutes
- If given, patient needs to be observed for at least 2 hours after the treatment prior to discharge
- Staff involved in giving this treatment need to wear appropriate PPE for aerosol generating procedure (Covid 19 precautions)

******Consider Alternate Diagnoses**

Such as bacterial tracheitis, Epiglottitis, Foreign body aspiration, Retro-pharyngeal or parapharyngeal abscess, Pneumonia, Bronchiolitis, or Chemical Inhalation **OR IF** Poor response to therapy, or age <6 months or > 6 years

Consider HNS consult for atypical presentations that do not respond to therapy

Treatments for Croup that are NOT Recommended due to lack of proven benefit:

- Antibiotics, Antitussives, Antihistamines, decongestants, inhaled or nebulized bronchodilators,
- Humidified air/mist in the office or inpatient settings⁵

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References:

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4. Bjornson C, Russell K, Vandermeer B, Klassen TP, Johnson DW. Nebulized epinephrine for croup in children. *Cochrane Database of Systematic Reviews* 2013, Issue 10. Art. No.: CD006199. DOI: 10.1002/14651858.CD006619.pub3.

5. Johnson DW. Croup. *BMJ Clin Evid*. 2014;2014:0321. Published 2014 Sep 29

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CROUP PATHWAY: EMERGENCY DEPARTMENT

Child 6 mos -6 years
Barky cough
And/or Stridor

****Consider alternate diagnoses:
No preceding URI symptoms
<6 mos or >6 years

Mild Croup

Barky Cough with any of the following:

-No stridor at rest

- Stridor only when agitated
- No respiratory distress
- No mental status changes
- No or minimal retractions
- No tachypnea

Moderate/Severe Croup

Stridor at rest with/without any of the following:

- Moderate to severe retractions
- Moderate to severe tachypnea
- Grunting, head bobbing, see-saw breathing
- Decreased air entry
- Anxious or agitated or tired appearing

Impending Respiratory Failure

- Toxic appearing
- Altered mental status
- Excessive drooling
- Airway obstruction
- Lethargy/Decreased level of consciousness
- Dusky or poor perfusion
- Cyanosis

****Dexamethasone Oral, IV or IM**
(0.6 mg/kg)
Max Dose 16 mg
*

Minimize agitation during evaluation and treatment
****One dose of Dexamethasone** in least invasive/ immediately available form
Oral, IV or IM (0.6 mg/kg) Max Dose 16 mg and
*****Nebulized Racemic Epinephrine 2.25% 0.5 mL**
Repeat every 30 mins until stridor at rest is resolved

*****Nebulized Racemic Epinephrine 2.25% 0.5 mL**
-Establish IV
-Non-rebreather mask
-Prepare to intubate if needed
****Decadron 0.6 mg/kg/dose IV**
-Consult PICU

Reassess

Clinically Improving?

Yes

Observe for minimum of 2 hours

No

Discharge home once discharge criteria is met

No

Recurrence of moderate to severe symptoms?

Yes

- ****Consider alternative diagnoses
- Consult Pediatrics/PICU

Discharge Criteria:

- No or minimal retractions
- No stridor at rest
- Tolerating orals
- Parents able to return pm
- 2 or fewer doses of Racemic epinephrine given after Dexamethasone

Not Routinely Recommended

- (No evidence suggesting the use of)
- Chest/Neck X-ray
- Antibiotics
- Viral Studies
- Cool mist in ED/hospital setting

Consider Inpatient Admission

- If a 3rd dose of Racemic Epinephrine is required
- Significant respiratory distress
- Stridor at rest 4-6 hours after post - Dexamethasone
- Poor oral intake
- Recurrent ED visits in a 24 hour period
- ***Suspicion of alternate diagnoses**
- Consider imaging if appropriate**
- ****Consider HNS consult**